Jiahao Tian

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Education

School of Mathematical Sciences, Zhejiang University — **Aug. 2023 – Jun. 2027 (Expected)**
Bachelor of Science in Statistics — *Hangzhou, Zhejiang, China*

- **GPA:** 88.31/100 or 3.98/4.30
- **Honors:** Selected for Chu Kochen Honors College (Physics); transferred to Statistics for strict mathematical training.

Selected Coursework

Probability Theory (92)	Stochastic Processes (99)	SDEs (Grad, 96)
Point Set Topology	Functional Analysis	Theory of Distributions
Mathematical Statistics (95)	Asymptotic Statistics (PhD Level)	

Research Experience

Robustness of Multiple-Knockoff Aggregation Methods — **Sept. 2025 – Present**
Undergraduate Researcher | Advised by Prof. Lijun Wang — *Hangzhou, China*

- Identified **catastrophic power collapse** (Power $\to 0$) in e-value, AKO, and voting-based aggregation under Cauchy distribution; characterized the **"zero-dilution effect"** where signal sparsity propagates across multiple runs.
- Proved a **multiplicity-based threshold inflation theorem** using **Extreme Value Theory (EVT)** and **Hoeffding's inequality**; showed that null multiplicity ($p - k \gg k$) forces thresholds to $\mathcal{O}(np)$, suppressing signals despite matched tail indices (1/2).
- Evaluated robust statistics (**LAD-Lasso**, **Cauchy Regression**, **Log-Squashing Lasso**) to mitigate infinite variance; analyzed how symmetric log-transforms stabilize heavy tails while preserving ordinal signal rankings.
- **First-author manuscript in preparation.**

Uncertainty Quantification via Conformal Prediction — **Mar. 2026 – Present**
Research Assistant | Advised by Prof. Guanhua Chen (UW-Madison) — *Remote*

- Established a **Fourier analytic framework** to prove the robustness of **Characteristic Function Distance (CFD)** kernels under extreme covariate shift; demonstrated that while Gaussian geometry collapses super-polynomially on sharp interfaces, **Laplace and Energy kernels** maintain stability via polynomial frequency penalization.
- Integrated CFD weighting into **UACQR** and **Split CP**, implementing a numerically stable pipeline with **Cholesky decomposition protection**; suppressed infinite interval rates (5.8%$\to$2.0%). Identified a **"dimensionality paradox"**: high-dimensional QRF self-corrects for shift via feature selection, while extreme weighting collapses **ESS** and induces variance–bias trade-off.

Academic Activities & Presentations

Personal Academic Space — **Spring 2026 – Present**
jhao1122.github.io/Notes — *Digital Repository*

- Systematized rigorous derivations for coursework in **SDEs**, **Real Analysis**, **Functional Analysis**, and **Asymptotic Statistics**.
- Archived technical notes and presentation materials for seminars on **Large Deviation Theory (LDP)** and **Derandomized Knockoff Methods**.

Talk: Convergence of Random Series and Large Deviations — **Fall 2025**
Advanced Probability Theory Seminar, Zhejiang University, 120 minutes — *Hangzhou, China*

Talk: Robustness of Multiple Knockoff Methods — **Fall 2025**
Weekly Statistical Seminar, Advised by Prof. Lijun Wang, 90 minutes — *Hangzhou, China*

Skills & Tests

Programming: R, C/C++, Julia, Python, MATLAB, Shell
Tools & OS: Quarto, Typst, Mkdocs, Linux, Git/GitHub, LaTeX
Independent Study: *Probability: Theory and Examples* (Durrett); *The Elements of Statistical Learning* (Hastie et al.)
Standardized Tests: GRE: Q170 | TOEFL: 5.5/6.0 (Old Scale:110/120)